Lab 4: Photovoltaics and Photodetectors

ESE 2180: Electronic, Photonic and Electromechanical Devices

Fall 2026

In this lab you will explore the characterization of photovoltaic devices and photoresistors (photocells) in dark and under illumination.

Part A: Measurement of Silicon Photovoltaic Devices

Background. Photons absorbed in a semiconductor generate electron-hole pairs. Carriers generated in or sufficiently near a PN junction can be collected, producing photocurrent. Under illumination, the solar cell can deliver electrical power to an external circuit. The dark I-V characteristic resembles that of a diode. The power-generating region lies at positive cell voltage and negative cell current under the diode sign convention.

You are provided with a calibrated solar cell whose open-circuit voltage (V_{OC}) and short-circuit current (I_{SC}) were measured by the manufacturer under standard illumination conditions of 1 sun ($100\,\mathrm{mW/cm^2}$). These values are labeled on the back of the solar cell.

In Section A of the lab, I-V characteristics are taken both in dark and under illumination from a lamp light source for Si photovoltaic devices. From the I-V data, one can determine the open-circuit voltage, short-circuit current, maximum power output, fill factor, and estimate the power conversion efficiency.

Materials

Si solar cell, source measurement unit, flashlight, and cables.

Procedures

1. To test your solar cell, connect the solar cell to the power supply and ammeter (in series). See Figure 1 for reference.
2. Manually measure the I-V characteristics in the dark by flipping the solar panel over.
    a. Connect the ammeter in series with the voltage source applied to your solar cell. Also, connect the voltmeter in parallel with the solar cell. Change the voltage supplied starting at 0 V with 50 mV step size and record the voltage across the solar cell and the current through the ammeter. Stop once you approach the current limit (about 20 mA). Now set the voltage source back to 0 V and switch the wires going into the voltage source so that you are supplying negative voltage to the cell. Take current readings from $-100\,\mathrm{mV}$ to $-600\,\mathrm{mV}$ with 100 mV step size.
3. Set your cell phone flashlight to its maximum brightness. Manually measure the I-V characteristics using the phone flashlight pointed at the solar cell. Use the same procedure as step

2a. Keep your phone in a fixed position throughout the measurement to ensure consistent illumination.

4. Without changing the position of the solar cell or flashlight, measure the short-circuit current (I_{SC}) and open-circuit voltage (V_{OC}) by short-circuiting and opening the circuit, respectively.

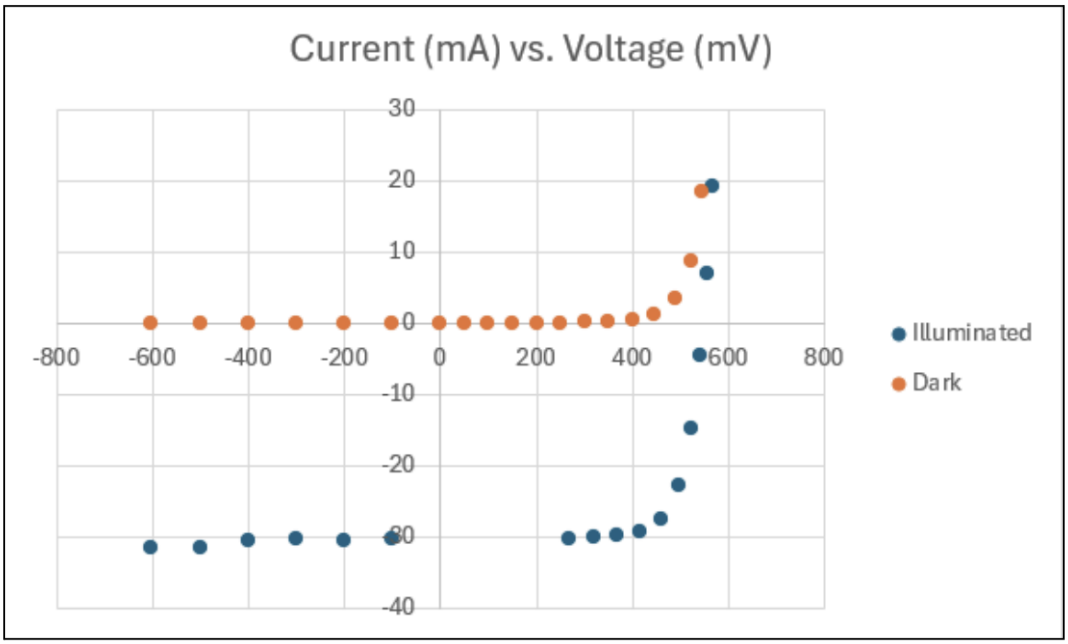
5. Reduce the brightness of your cell phone flashlight. Measure the I_{SC} and V_{OC} at two additional brightness levels. Note that you don't need to measure the full I-V curve at each brightness. Simply measure I_{SC} and V_{OC} at each level. Keep the positions of the flashlight and solar cell unchanged throughout the measurements.

6. Record I_{SC} and V_{OC} written on the backside of your solar cell.

Part A: Data Processing, Discussion, and Analysis

Answer the following questions with short responses, max 3–4 sentences. Show calculations where requested.

1) Plot dark and flashlight I-V curves on the same axes. Label the axes and identify the fourth-quadrant power-generating region. Use the measured voltage across the solar cell, not the voltage set on the DC power supply, for the horizontal axis. **(1 pt)**



X-Axis: Voltage (mV)

Y-Axis: Current (mA)

There is a gap in the forward bias voltages for the illuminated condition because it was not possible to set the power supply voltage below 268 mV (due to the voltage generated by the photovoltaic cell).

The power generating region is the bottom right quadrant, where there are only values from the illuminated cell.

2) Let's determine the output power. For illuminated conditions, determine I_{SC}, V_{OC}, voltage (V_M) and current (I_M) at the maximum power point. Calculate the maximum electrical power output, $P_{max} = |V_M I_M|$. **(0.5 pt)**

$$\begin{aligned} I_{SC} &= -26.165\,\text{mA}, \\ V_{OC} &= 545\,\text{mV}, \\ I_M &= -22.707\,\text{mA}, \\ V_M &= 495\,\text{mV}. \end{aligned}$$

$$\begin{aligned} P_{max} &= |(-22.707\,\text{mA})(0.495\,\text{V})| \\ &= 11.239965\,\text{mW}. \end{aligned}$$

3) Let's estimate the input power.

(a) What are your solar cell's V_{OC} and I_{SC} under 1 sun ($100\,\text{mW/cm}^2$)? Check the back side of your solar cell. **(0.25 pt)**

$$V_{OC} = 0.6\,\text{V}$$
$$I_{SC} = 312\,\text{mA} = 0.312\,\text{A}$$

(b) Since the short-circuit current is approximately proportional to the incident optical power (check this out if you are interested), you can estimate the optical power delivered by the flashlight to the solar cell by comparing the short-circuit current measured under flashlight illumination with the manufacturer's reference value measured under 1 sun. The dimensions of your solar cell are either 2 cm by 4 cm or 2 cm by 2 cm. Be careful with the units when you calculate the input power. **(0.5 pt)**

$$A = (4\,\text{cm})(2\,\text{cm}) = 8\,\text{cm}^2.$$

I_{SC} in our conditions: $-26.165\,\text{mA}$. I'm using the absolute value of I_{SC} for the calculations.

$$\frac{P_{in}/(8\,\text{cm}^2)}{26.165\,\text{mA}} = \frac{100\,\text{mW/cm}^2}{312\,\text{mA}}.$$

$$P_{in} = 67.077\,\text{mW}.$$

(c) Explain why this method provides only an estimate of the actual optical irradiance rather than a calibrated measurement. In your explanation, consider the differences between the emission spectrum of your cell phone flashlight (which you observed in Lab 3) and the solar spectrum. **(0.25 pt)**

This method only provides only an estimate of the actual optical irradiance rather than a calibrated measurement of the solar panel since a white phone light has concentrated spectral features, to which sunlight spans a broader range of wavelengths. This would directly cause altered power generation as silicon's responsivity depends on wavelength, so scaling the one-sun current therefore gives only an estimate, and differences in illumination uniformity, also cell temperature introduces additional uncertainty.

4) Calculate the efficiency and fill factor of your solar cell. **(0.5 pt)**

$$\begin{aligned}\text{Efficiency} &= \left|\frac{P_{\max}}{P_{\text{in}}}\right| \times 100\% \\ &= \frac{11.23\,\text{mW}}{67.077\,\text{mW}} \times 100\% \\ &= 16.74\%.\end{aligned}$$

$$\begin{aligned}\text{Fill Factor} &= \frac{V_M I_M}{V_{\text{OC}} I_{\text{SC}}} \\ &= \frac{11.23\,\text{mW}}{(26.165\,\text{mA})(0.545\,\text{V})} \\ &= 0.7875.\end{aligned}$$

5) How did your I_{SC} and V_{OC} readings change as you reduced the brightness of your cell phone flashlight? Explain why I_{SC} and V_{OC} respond differently to decreasing illumination intensity. Which is approximately linear, and which is approximately logarithmic? **(1 pt)**

When we lowered the flashlight brightness, I_{SC} dropped a lot, from 26.2 mA to 14.3 mA to 5.47 mA (about 79% lower). V_{OC} only dropped a little, from 545 mV to 526 mV to 495 mV (about 9% lower). I_{SC} is roughly linear with light intensity, because more light makes more electron-hole pairs, so more current is collected. V_{OC} is roughly logarithmic, because it follows

$$V_{\text{OC}} \approx \frac{kT}{q} \ln\left(\frac{I_{\text{SC}}}{I_0}\right),$$

so a big change in light causes only a small change in voltage.

6) What area of the solar cell would be needed to produce 5 kilowatts of power? (Assume the voltage produced is constant and that the current would be proportional to the area of the solar cell). As a reference, 5 kilowatts is the power required to supply heat to a typical single-family house in the US. **(1 pt)**

Under the flashlight,

$$P_{\max} = |(-22.707\,\text{mA})(0.495\,\text{V})| = 11.24\,\text{mW}$$

for a cell area of $8\,\text{cm}^2$.

The power per area is

$$\frac{11.24\,\text{mW}}{8\,\text{cm}^2} = 1.405\,\text{mW/cm}^2.$$

Since power is proportional to area, the area needed is

$$\begin{aligned}A_{\text{needed}} &= \frac{5{,}000{,}000\,\text{mW}}{1.405\,\text{mW/cm}^2} \\ &\approx 3.56 \times 10^6\,\text{cm}^2 \\ &\approx 356\,\text{m}^2.\end{aligned}$$

7) **(a)** Check what you learned during the lecture. Suggest two ways to increase electrical power output. It is important to justify your answer. **(0.5 pt)**

Power output can be increased by increasing the incident irradiance on the solar panel since increased absorbed photons equate to more electron-hole pairs which increase photocurrent in the solar panel. Power output can also be increased via increasing the surface area of the solar panel, as a larger area equates to more photons to be absorbed.

(b) Explain whether each necessarily improves power conversion efficiency, $\eta = P_{\max}/P_{\text{in}}$. **(0.5 pt)**

Neither change necessarily increases $\eta = P_{\max}/P_{\text{in}}$. Increasing area scales electrical output and optical input by the same factor, and so it leaves efficiency unchanged. Also, increasing irradiance increases optical input, but V_{OC} rises logarithmically, heating and resistive losses can offset gains, so a larger $P_{\max}$ alone does not establish improved efficiency.

(c) Suggest one materials/device-design change that could improve η, and connect it to absorption, recombination/collection, voltage, or fill factor. **(1 pt)**

One way to improve efficiency is to use silicon with fewer crystal defects. Defects can cause photogenerated electrons and holes to recombine before they are collected. Reducing these defects allows more carriers to reach the junction, increasing photocurrent and electrical output for the same incident optical power.

Part B: Photoresistors

In this part we are going to measure simple I-V characteristics of a CdS photoresistor (colloquially known as "Photocell" and also known as photoconductor).

Materials

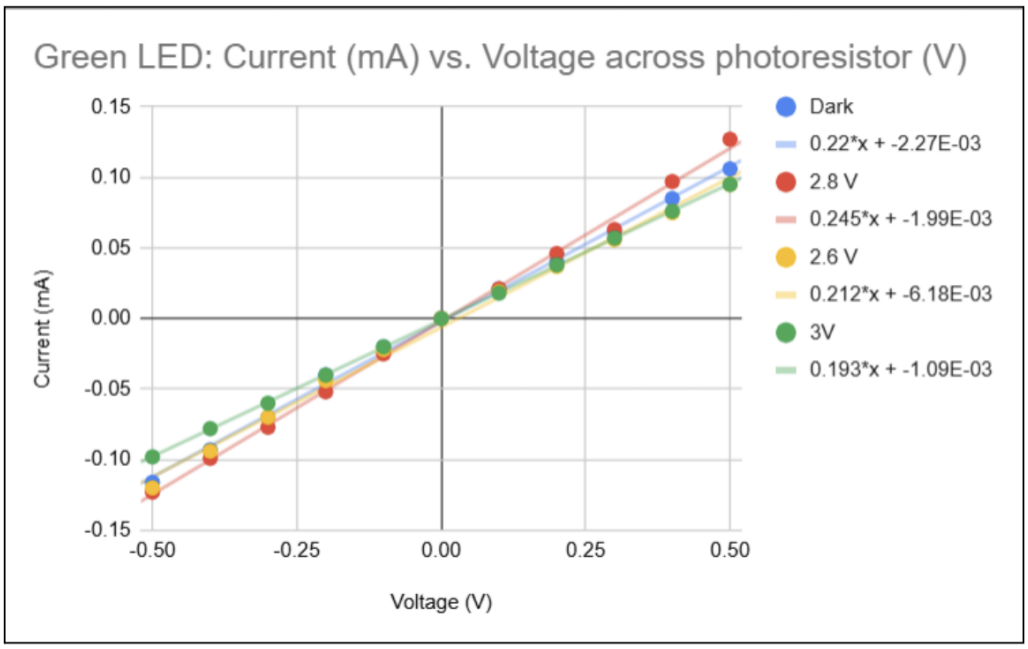
CdS photoresistor, R, G, B LEDs, source measurement unit.

Procedures

1. To test your device, connect the wires to the right colored terminals and mount them on the stand facing normal to the illumination pathway.
2. For the photoresistor, first measure I-V curves in the dark manually by changing the voltage supplied to the photoresistor from $-0.5\,\text{V}$ to $0.5\,\text{V}$ at a step size of $0.1\,\text{V}$ and reading off the current flowing through the photoresistor. Total of 11 data points.
3. Then turn on the green LED at a voltage of $2.8\,\text{V}$ ($1.52\,\text{mW}$) of power supply. Change the voltage supplied to the photoresistor from $-0.5\,\text{V}$ to $0.5\,\text{V}$ at a step size of $0.1\,\text{V}$ and read off the current flowing through the photoresistor. Total of 11 data points.
4. Repeat step 3 with the green LED at a voltage of $2.6\,\text{V}$ ($0.73\,\text{mW}$) and $3\,\text{V}$ ($2\,\text{mW}$).
5. Repeat step 3 for red ($1.83\,\text{V}$) and blue ($2.70\,\text{V}$) LEDs.

Part B: Data Processing, Discussion, and Analysis

1) Plot the I-V curves for dark and under three different green illuminations on the same plot. From the line of best fit for each curve, linearly extrapolate the resistance. Tabulate these values. **(0.5 pts)**



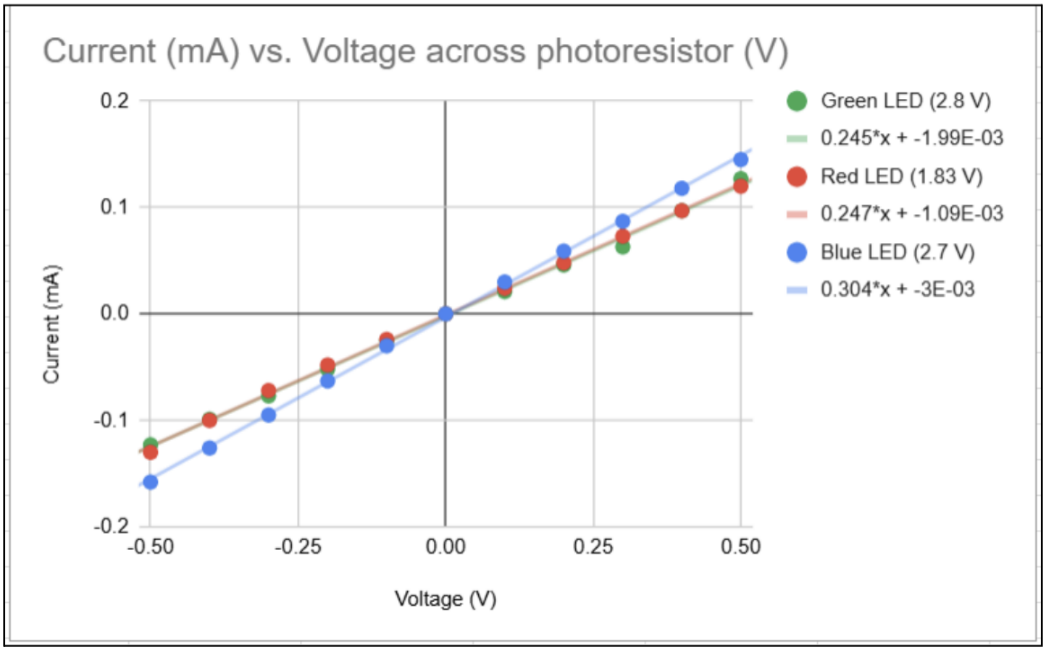
Green LED Voltage (V)	Resistance (Ω)
Dark	4540
2.6	4716
2.8	4080
3	5180

Line of best fit gives slope as I/V, but $V = IR$, $R = V/I$, so resistance is 1/slope. The units that we graphed are V and mA, so 1/slope will give us resistance in kΩ.

$$R_{\text{dark}} = \frac{1}{0.22} = 4.54\,\text{k}\Omega,$$
$$R_{2.8} = \frac{1}{0.245} = 4.08\,\text{k}\Omega,$$
$$R_{2.6} = \frac{1}{0.212} = 4.716\,\text{k}\Omega,$$
$$R_{3} = \frac{1}{0.193} = 5.18\,\text{k}\Omega.$$

Our measurement for Green LED at 2.8 V is an outlier as it does not follow our established relationship as LED voltage increases, so does the resistance, possible lab errors include inconsistent illumination, settling time, or contact resistance could contribute.

2) Plot the I-V curves for dark and the three different color LEDs (for Green, use 2.8 V dataset). From the line of best fit for each curve, linearly extrapolate the resistance. Tabulate these values. **(0.5 pts)**



LED	Resistance (Ω)
Dark	4545
Red (1.83 V)	4049
Green (2.8 V)	4081
Blue (2.70 V)	3289

3) **(a)** At an applied voltage of 0.5 V, calculate the dark current and illuminated current for the green LED. Determine the photocurrent and the ratio of photocurrent to dark current. **(0.5 pt)**

At 0.5 V, the dark current is 0.106 mA and the illuminated current is 0.127 mA. The photocurrent is

$$I_{\text{photo}} = 0.127 - 0.106 = 0.021\,\text{mA}.$$

The ratio of photocurrent to dark current is

$$\frac{I_{\text{photo}}}{I_{\text{dark}}} = \frac{0.021}{0.106} \approx 0.20.$$

(b) Why is a large photocurrent-to-dark-current ratio desirable for a photodetector? **(0.5 pt)**

A large photocurrent-to-dark-current ratio is desirable for a photodetector because it makes the light-induced change easier to distinguish from the dark background and its drift. The photodetector will be more sensitive, because photocurrent needs to be greater than dark current for it to be detected, and larger changes in photocurrent makes changes in intensity easier to detect.

4) **(a)** Explain how increasing illumination intensity affects the resistance of a CdS photoresistor. Relate your answer to photogenerated carrier concentration and conductivity. **(0.5 pt)**

CdS is an n-type semiconductor. So, increasing illumination intensity increases carrier concentration, which decreases resistance (dominant term is $\mu q(n + n_0)$), because the photogenerated carrier concentration increases conductivity.

(b) The bandgap of CdS is approximately 2.42 eV. Calculate the corresponding cutoff wavelength. Based on this value, discuss the expected response of CdS to red, green, and blue illumination. How do your experimental results compare with your prediction? **(0.5 pt)**

$$E_g = \frac{hc}{\lambda}$$

$$\lambda = \frac{1240\,\text{eV nm}}{2.4\,\text{eV}} = 517\,\text{nm}.$$

The cutoff is 517 nm, which corresponds to blue-green light. Blue photons with wavelengths below 517 nm can generate band-to-band carriers, to which red photons and green photons above this cutoff should create a weaker intrinsic response. Green response depends on the LED spectrum relative to the absorption edge, and defects can allow a response below the bandgap.

Experimentally, blue gives the lowest resistance (3289 Ω), while red (4049 Ω) and green (4081 Ω) are similar and both below dark (4545 Ω), which was consistent with the strongest response under blue but with a measurable red/green response. Because the incident powers and spectra of the three colors were not normalized, these resistances alone do not establish intrinsic spectral responsivity.